

Kernel-Matched Continuum Attention for Looped Gaussian-Process Inference: Scaling Laws and Discretization-Covariant Posterior Geometry

Janis

Hancya

Abstract

We develop a theory of nonlinear looped attention for Gaussian-process and kernel least-squares regression in function space. The organizing principle is that the attention nonlinearity is prescribed by the covariance kernel: it is a modelling choice, not a temperature learned from data. Exponential softmax scores realize normalized strictly positive kernels, including the Gaussian radial basis kernel after a norm correction; signed positive semidefinite kernels instead require an explicit Mercer feature map. In both cases quadrature weights are part of attention, so the discrete layer approximates a continuum integral operator rather than the sampling density.

The nonlinear attention block extracts a prompt-dependent finite-dimensional subspace, while a hard-coded Ritz construction turns that subspace into a symmetric positive-definite approximation of the posterior covariance. A tied Heavy-Ball or nonstationary Chebyshev recurrence then computes the posterior mean using exact matrix-vector products. We prove continuum consistency, covariance across discretizations, positivity of the learned metric, and mesh-uniform convergence whenever the prompt-conditioned subspace captures the data-informed posterior directions uniformly. In a commuting Gaussian sequence model we obtain an exact modal risk formula resolving depth, effective width, context, training time, and discretization. For Mercer eigenvalues $\lambda_j \asymp j^{-a}$, the context contribution is $n^{-(a-1)/a}$, the rank contribution is S^{1-a} , and the loop contribution is the appropriate Heavy-Ball or Chebyshev polynomial residual. This lifts the fixed- and randomly-rotated-spectrum dichotomy of linear in-context regression to kernel posterior operators. We finally specialize the construction to linear Bayesian inverse problems and to local Gauss-Newton-Laplace subproblems for nonlinear PDEs. No numerical algebra is learned: attention learns the posterior geometry, and exact looping dynamics perform inference. A failure-preserving validation campaign verifies the spectral certificates, reduced RMT/DMFT scaling laws, and nonuniform-mesh transfer. On a variable-coefficient elliptic inverse problem with $N = 16384$ unknowns and a low-rank-plus-floor latent covariance, the setup-inclusive one-query time at context length $m = 4096$ is 10.90 ms for kernel-Ritz Heavy-Ball, 24.03 ms for exact Woodbury, and 139.79 ms for dense Cholesky on an H100. Exact Woodbury remains faster at $m = 512$, so the speed claim is explicitly restricted to the measured long-context regime.

1 Introduction

In-context regression and Bayesian inversion both reduce, after the relevant features have been chosen, to applying an inverse covariance or precision operator. This observation has motivated interpretations of linear self-attention as gradient descent or preconditioned least squares [1, 4, 12] and a recent theory of how depth, width, context, and optimization time interact in in-context linear regression [2]. The linear model is deliberately transparent, but it leaves open two questions that are essential for Gaussian-process (GP) regression and inverse PDEs. Which nonlinearity should replace the Euclidean covariance contraction, and how can the resulting model represent one operator independently of the points or mesh on which its context happens to be observed?

Our answer separates modelling from learning. A GP is Gaussian because each finite collection of evaluations is jointly Gaussian; its covariance kernel need not be a Gaussian radial basis

function [9]. The covariance kernel is selected to encode regularity, locality, periodicity, or a PDE Green function. We prescribe an attention interaction compatible with that kernel. For a strictly positive kernel κ , normalized kernel attention is obtained from the fixed nonlinearity $\sigma_\kappa(s) = \exp(s)$ and score $s = \log \kappa$. For the RBF kernel this is ordinary softmax with a fixed length scale and an explicit norm bias. For Matérn, spectral, Green, or signed kernels, the appropriate interaction may instead be a deterministic Mercer, Fourier, or operator feature map. There is no universal softmax claim and no learned temperature in our theory.

This distinction is also needed algebraically. Row-normalized softmax is a Markov operator; it is not in general a symmetric posterior covariance. Consequently we use nonlinear attention only to extract task-dependent feature fields. Orthonormalization in the continuum mass metric followed by an exact Ritz formula produces a positive-definite posterior metric. This metric is held fixed while a recurrent, weight-tied polynomial method solves the regularized normal equations. The architecture therefore has two auditable parts:

kernel-matched continuum attention learns the geometry

↓

exact Heavy-Ball/Chebyshev dynamics perform inference.

The continuum formulation matters. Neural operators already share parameters across discretizations [6], and continuum attention has been formulated as an integral operator whose usual discrete implementation is a Monte Carlo or quadrature approximation [3]. Learned preconditioners can also transfer between regular and irregular PDE meshes [7]. Our target is more specific: infer from heterogeneous function observations a right-hand-side-independent approximation of the *posterior covariance* of an inverse problem, and couple it to an exact loop with a convergence rate uniform in the discretization. Classical finite elements and multigrid are not claimed to fail on irregular meshes; rather, the statistical object being amortized here is task-specific posterior geometry.

Contributions.

1. We define kernel-matched continuum attention with quadrature weights and show exactly when softmax realizes the desired normalized kernel. We characterize its reversible self-adjoint representation and explain why positive definiteness of posterior geometry must be imposed separately.
2. We construct a prompt-conditioned Ritz metric which is symmetric positive definite by design and equals the posterior covariance whenever the learned subspace contains all data-informed directions.
3. We prove a discretization-covariance estimate and mesh-uniform spectral equivalence. These imply resolution-independent Heavy-Ball and Chebyshev rates with the same coefficients at every mesh size.
4. We give an exact Gaussian sequence risk formula that separates context n , effective width S , loop depth L , training time t , and discretization h . Power-law Mercer spectra yield explicit scaling laws.
5. We lift the fixed-spectrum/randomly-rotated-spectrum distinction to posterior operators and show why prompt conditioning, rather than a global preconditioner, is necessary when eigenspaces vary across tasks.
6. We validate the complete chain—kernel quadrature, Ritz spectral certificates, linear-Wishart controls, kernel-specific resolvents, reduced DMFT exponents, discretization covariance, and elliptic posterior solves— while retaining the regime in which exact Woodbury is the faster baseline.

All deterministic statements below are proved. The spectral scaling laws are theorems for the stated commuting sequence model; they are not presented as a replica prediction for an unspecified nonlinear Transformer. This boundary keeps the nonlinear kernel choice and the exact numerical dynamics distinct from assumptions on how features are learned. The numerical campaign mirrors that boundary: Marchenko–Pastur is tested only on a linear-Wishart control, whereas the nonlinear RBF operator is evaluated through its own empirical one- and two-resolvent statistics.

2 Bayesian regression in whitened coordinates

Let $\mathcal{U}$ and $\mathcal{Y}$ be real separable Hilbert spaces. Let $C_0 : \mathcal{U} \rightarrow \mathcal{U}$ be a self-adjoint, positive, injective trace-class operator, and let $\Gamma_\varepsilon : \mathcal{Y} \rightarrow \mathcal{Y}$ be self-adjoint with $\Gamma_\varepsilon \succeq \gamma_\varepsilon I$ for some $\gamma_\varepsilon > 0$. Consider

$$m \sim \mathcal{N}(0, C_0), \quad y = Gm + \eta, \quad \eta \sim \mathcal{N}(0, \Gamma_\varepsilon), \quad (2.1)$$

where $G : \mathcal{U} \rightarrow \mathcal{Y}$ is bounded. The covariance of the observations, $GC_0G^* + \Gamma_\varepsilon$, the posterior precision, and the posterior covariance are different objects. The distinction is easiest after prior whitening. Set

$$A = C_0^{1/2}G^*\Gamma_\varepsilon^{-1}GC_0^{1/2}, \quad H = I + A, \quad g = C_0^{1/2}G^*\Gamma_\varepsilon^{-1}y. \quad (2.2)$$

Then $A \succeq 0$, and the posterior mean and covariance are

$$m_{\text{post}} = C_0^{1/2}H^{-1}g, \quad C_{\text{post}} = C_0^{1/2}H^{-1}C_0^{1/2}. \quad (2.3)$$

The operator learned by our attention head is an approximation $B_\theta(\mathcal{C}) \simeq H^{-1}$ in whitened coordinates. It is not simply the inverse empirical covariance of the input tokens.

Remark 2.1 (GP regression and kernel ridge regression). If $\mathcal{U}$ is the Cameron–Martin coordinate space of a GP with covariance kernel k_0 and G is an evaluation operator, then (2.3) is GP regression. The posterior mean is also the kernel ridge minimizer with regularization determined by the observation noise [9, 13]. None of the results below requires k_0 to be RBF.

3 The nonlinearity is prescribed by the kernel

Let (X, ρ) be a compact measured space, let $\kappa : X \times X \rightarrow \mathbb{R}$ be continuous, and suppose first that $\kappa(x, y) > 0$. Define

$$d_\kappa(x) = \int_X \kappa(x, y) \rho(dy), \quad (\mathcal{A}_\kappa v)(x) = \frac{1}{d_\kappa(x)} \int_X \kappa(x, y)v(y) \rho(dy). \quad (3.1)$$

This is a normalized kernel integral operator. Given nodes $X_h = \{x_j\}_{j=1}^{N_h}$ and positive quadrature weights ω_j , its discrete counterpart is

$$(A_{\kappa, h}v)_i = \sum_{j=1}^{N_h} a_{ij}^{\kappa, h} v_j, \quad a_{ij}^{\kappa, h} = \frac{\omega_j \kappa(x_i, x_j)}{\sum_{\ell=1}^{N_h} \omega_\ell \kappa(x_i, x_\ell)}. \quad (3.2)$$

Equivalently, set the fixed score $s_\kappa(x, y) = \log \kappa(x, y)$ and apply softmax to $s_\kappa(x_i, x_j) + \log \omega_j$. The $\log \omega_j$ term is not a positional heuristic; it makes the sum approximate integration against ρ . Omitting it changes the continuum measure when the sampling density changes.

Example 3.1 (RBF softmax without a learned temperature). For

$$\kappa_\ell(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\ell^2}\right),$$

ordinary softmax with logits

$$\frac{x_i^\top x_j}{\ell^2} - \frac{\|x_j\|^2}{2\ell^2} + \log \omega_j$$

gives (3.2), because the omitted term $-\|x_i\|^2/(2\ell^2)$ is constant along row i . The scale ℓ is part of the GP prior or data model. It is fixed before training in the present theory.

Remark 3.2 (What softmax cannot do). A GP covariance may be Matérn, periodic, polynomial, signed pointwise, or a Green kernel. A row-stochastic positive matrix cannot equal an arbitrary covariance matrix. For a general positive-semidefinite kernel we instead use a prescribed feature map φ_κ satisfying $\kappa(x, y) = \langle \varphi_\kappa(x), \varphi_\kappa(y) \rangle$ and implement the unnormalized contraction by linear attention [5]. The term *kernel-matched attention* refers jointly to the feature map, score nonlinearity, normalization, and quadrature measure.

Although $\mathcal{A}_\kappa$ need not be self-adjoint in $L^2(\rho)$, it is reversible. Let

$$Z_\kappa = \int_X d_\kappa(x) \rho(dx), \quad \pi_\kappa(dx) = Z_\kappa^{-1} d_\kappa(x) \rho(dx). \quad (3.3)$$

Proposition 3.3 (Reversibility and positivity). *If κ is symmetric, then $\mathcal{A}_\kappa$ is self-adjoint on $L^2(\pi_\kappa)$. If κ is additionally positive semidefinite as an integral kernel, then $\mathcal{A}_\kappa \succeq 0$ on $L^2(\pi_\kappa)$. The discrete matrix $A_{\kappa,h}$ has the analogous properties for masses $\pi_{h,i} \propto \omega_i d_{\kappa,h}(x_i)$.*

Assumption 3.4 (Quadrature class). *There is a normed class $\mathcal{Q}$, an order $p > 0$, and uniformly bounded sampling maps P_h such that, for every $x \in X$ and every value field v under consideration,*

$$\begin{aligned} \left| \int_X \kappa(x, y) \rho(dy) - \sum_j \omega_j \kappa(x, x_j) \right| &\leq C_0 h^p, \\ \left| \int_X \kappa(x, y) v(y) \rho(dy) - \sum_j \omega_j \kappa(x, x_j) v(x_j) \right| &\leq C_v h^p. \end{aligned} \quad (3.4)$$

Moreover $d_\kappa(x) \geq d_0 > 0$.

Theorem 3.5 (Continuum consistency). *Under Assumption 3.4,*

$$\|A_{\kappa,h} P_h v - P_h \mathcal{A}_\kappa v\|_{\ell^\infty} \leq C(1 + \|v\|_\infty) h^p, \quad (3.5)$$

with C independent of the locations of the nodes within the admissible quadrature family. In particular, the same parameters define one continuum operator on nonuniform discretizations.

The layer is nonlinear as a function of its queries, keys, and prompt, even though it is linear in the value channel after the attention weights have been fixed. Our statistical head may mix the resulting contextual fields, but it cannot change σ_κ or the kernel hyperparameters. Empirical Bayes or deep-kernel learning would be a different model class.

4 A prompt-conditioned posterior metric

Let the kernel-matched encoder applied to a prompt $\mathcal{C}$ return S fields $\phi_{\theta,1}(\mathcal{C}), \dots, \phi_{\theta,S}(\mathcal{C}) \in \mathcal{U}$. A fixed Gram–Schmidt or polar factor in the $\mathcal{U}$ inner product gives an isometry $Q : \mathbb{R}^S \rightarrow \mathcal{U}$, $Q^*Q = I$. Write $P = QQ^*$. Define

$$B_Q = I - P + Q(Q^*HQ)^{-1}Q^*. \quad (4.1)$$

Only the range of Q is learned. Projection, the $S \times S$ Galerkin matrix, its Cholesky factorization, and every application of B_Q are exact algebraic operations.

Proposition 4.1 (Posterior geometry by construction). *The operator B_Q is self-adjoint, positive definite, and satisfies $0 \prec B_Q \preceq I$. If $\text{range}(A) \subseteq \text{range}(Q)$, then $B_Q = H^{-1}$. More generally,*

$$B_Q = I - Q \left[I - (Q^* H Q)^{-1} \right] Q^*, \quad (4.2)$$

so B_Q is the prior-whitened covariance minus a positive finite-rank data correction.

This sign is forced by Bayesian conditioning. An arbitrary positive low-rank addition to the prior would have the wrong geometry.

Let $Ae_j = \nu_j e_j$ with $\nu_1 \geq \nu_2 \geq \dots \geq 0$, and let P_S^* project onto $\text{span}(e_1, \dots, e_S)$. Put

$$r_S = \sup_{j>S} \frac{\nu_j}{1 + \nu_j}, \quad M = \|H\| = 1 + \nu_1, \quad C_M = 1 + 2\sqrt{2} + 2\sqrt{2}M. \quad (4.3)$$

Theorem 4.2 (Subspace error controls the effective spectrum). *If $\delta_S = \|P - P_S^*\|_{\text{op}}$, then*

$$\|B_Q - H^{-1}\|_{\text{op}} \leq r_S + C_M \delta_S. \quad (4.4)$$

If

$$\epsilon_S = M(r_S + C_M \delta_S) < 1, \quad (4.5)$$

then

$$\text{spec}(B_Q^{1/2} H B_Q^{1/2}) \subset [1 - \epsilon_S, 1 + \epsilon_S], \quad \text{cond}(B_Q^{1/2} H B_Q^{1/2}) \leq \frac{1 + \epsilon_S}{1 - \epsilon_S}. \quad (4.6)$$

The theorem reveals the role of effective width. Increasing S reduces the spectral tail r_S ; increasing context or training time can reduce the principal-angle error δ_S . Width alone is not sufficient when the prompt-dependent eigenspace has not been identified.

5 Covariance across discretizations

Let $\mathcal{U}_h$ be a finite-dimensional trial space with mass matrix M_h and inner product $\langle u, v \rangle_h = u^\top M_h v$. All QR factorizations and adjoints are taken in this metric. Let $\iota_h : \mathcal{U}_h \rightarrow \mathcal{U}$ be an isometric reconstruction after mass whitening and let $\Pi_h = \iota_h \iota_h^*$. For comparison on a common space, define the lifted operators

$$\bar{H}_h = \iota_h H_h \iota_h^* + (I - \Pi_h), \quad \bar{P}_h = \iota_h Q_h Q_h^* \iota_h^*. \quad (5.1)$$

Assumption 5.1 (Consistent posterior and feature fields). *For a continuum posterior precision H and a continuum feature projector P , there are constants c_H, c_P and $p > 0$ such that*

$$\|\bar{H}_h - H\|_{\text{op}} \leq c_H h^p, \quad \|\bar{P}_h - P\|_{\text{op}} \leq c_P h^p + \varepsilon_{\text{enc}}, \quad (5.2)$$

uniformly over the admissible prompts and discretizations. Also $I \preceq H, \bar{H}_h \preceq M I$.

The encoder term ε_{enc} is statistical; the powers of h arise from quadrature and trial-space approximation. Theorem 3.5 supplies the attention component of the second bound when the remaining pointwise mixing and orthonormalization maps are uniformly Lipschitz away from rank loss.

Theorem 5.2 (Discretization-covariant Ritz metric). *Under Assumption 5.1, let $\bar{B}_h$ be the lift of the discrete metric (4.1) and let B be its continuum analogue. Then*

$$\|\bar{B}_h - B\|_{\text{op}} \leq C_M ((c_H + c_P) h^p + \varepsilon_{\text{enc}}). \quad (5.3)$$

For the transfer $I_{h \rightarrow h'} = \iota_{h'}^* \iota_h$,

$$\|I_{h \rightarrow h'} B_h - B_{h'} I_{h \rightarrow h'}\|_{\text{op}} \leq C_M [(c_H + c_P)(h^p + h'^p) + 2\varepsilon_{\text{enc}}] + \varepsilon_{\text{tr}}, \quad (5.4)$$

where ε_{tr} is the consistency error caused by projecting between non-nested trial spaces and vanishes for exact lifted comparison.

Corollary 5.3 (Mesh-independent effective condition number). *Suppose a continuum metric satisfies $\text{spec}(B^{1/2}HB^{1/2}) \subset [\mu_0, L_0]$ with $\mu_0 > 0$, and suppose the encoder floor ε_{enc} is small enough that the induced symmetrically preconditioned perturbation is strictly below $\mu_0/2$. For all sufficiently small h ,*

$$\text{spec}(B_h^{1/2}H_hB_h^{1/2}) \subset [\mu, L], \quad (5.5)$$

where $0 < \mu < L < \infty$ are independent of h .

Accepting variable sequence length is weaker than (5.4). The latter states that the learned object is a single continuum operator up to controlled discretization and encoder error.

6 Exact looped inference

We now hold $B = B_Q$ fixed for one prompt and solve

$$Hx = g. \quad (6.1)$$

The Hessian–vector product $v \mapsto Hv$ is an exact unnormalized bilinear contraction; in a least-squares problem it is

$$v \mapsto v + C_0^{1/2}G^*\Gamma_\varepsilon^{-1}GC_0^{1/2}v.$$

It should not be replaced by row-normalized softmax. Kernel-matched nonlinear attention constructs B ; linear attention or a weak-form operator token applies H .

6.1 Heavy–Ball as a tied two-state block

With $x_{-1} = x_0 = 0$, define

$$x_{\ell+1} = x_\ell + \alpha B(g - Hx_\ell) + \beta(x_\ell - x_{\ell-1}). \quad (6.2)$$

The same block is used at every depth. It stores one delayed state, applies one Hessian–vector product and one factorized metric, and uses no adaptive inner products or divisions [8].

Let $K = B^{1/2}HB^{1/2}$ and suppose $\text{spec}(K) \subset [\mu, L]$. The error polynomials satisfy

$$p_{-1}(z) = p_0(z) = 1, \quad p_{\ell+1}(z) = (1 + \beta - \alpha z)p_\ell(z) - \beta p_{\ell-1}(z). \quad (6.3)$$

Theorem 6.1 (Finite-depth Heavy–Ball bound). *Set*

$$\alpha_\star = \frac{4}{(\sqrt{L} + \sqrt{\mu})^2}, \quad \beta_\star = \left(\frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}} \right)^2, \quad q = \sqrt{\beta_\star}. \quad (6.4)$$

Then, for every $\ell \geq 0$,

$$\|x_\ell - x_\star\|_{B^{-1}} \leq [1 + \ell(1 + q)]q^\ell \|x_0 - x_\star\|_{B^{-1}}. \quad (6.5)$$

Constant coefficients are strictly stable on $[\mu, L]$ whenever

$$0 \leq \beta < 1, \quad 0 < \alpha L < 2(1 + \beta). \quad (6.6)$$

The polynomial prefactor in (6.5) cannot in general be removed: the characteristic roots coalesce at the spectral endpoints.

6.2 Chebyshev and PCG

Among polynomials of degree at most ℓ with $p(0) = 1$, the scaled Chebyshev residual satisfies

$$\max_{z \in [\mu, L]} |p_\ell^{\text{Ch}}(z)| = \frac{1}{\left| T_\ell \left(\frac{L+\mu}{L-\mu} \right) \right|} \leq 2q^\ell. \quad (6.7)$$

Its coefficients vary with ℓ but can be generated by a hard-coded scalar recurrence, so vector weights remain tied. Chebyshev is the minimax predetermined polynomial for a certified interval. Preconditioned conjugate gradients (PCG) chooses its polynomial from the realized right-hand side and Krylov history. PCG is therefore the stronger adaptive numerical reference, but it requires reductions, guarded divisions, and more recurrent state [10]. We make no universal domination claim over PCG.

Proposition 6.2 (Exact fixed-polynomial Bayes risk). *Let $c = B^{1/2}g$ have conditional covariance Σ_c , let $K = B^{1/2}HB^{1/2}$, and let a fixed polynomial method have residual p_ℓ . Then*

$$\mathbb{E}[\|x_\ell - x_\star\|_H^2 \mid H, B] = \text{tr}(\Sigma_c K^{-1} p_\ell(K)^2) = \int \frac{p_\ell(z)^2}{z} \nu_c(dz), \quad (6.8)$$

where ν_c is the Σ_c -weighted spectral measure of K . For PCG the residual polynomial depends on c , so the last deterministic spectral substitution is not valid without a joint Krylov state evolution.

Equations (6.3) and (6.8) are the precise entry point for random-matrix or dynamical mean-field limits: once a limiting weighted spectral measure has been derived, every fixed-depth HB or Chebyshev risk follows by integration. The nonlinear kernel affects that measure; it does not alter the exact polynomial algebra.

7 Direct lift of the reduced preconditioner scaling calculus

We now make the correspondence with the reduced recurrent model of Bordelon et al. [2] explicit. This section uses L_d for recurrent depth. Let $T \succeq 0$ be a compact kernel covariance operator and let $\Omega \succeq 0$ be the covariance of regression tasks. In the long-context population limit, the operator analogue of their reduced loss is

$$\mathcal{L}_{L_d}(\Gamma; T, \Omega) = \text{tr}\{\Omega R_{L_d}(\Gamma, T)^* T R_{L_d}(\Gamma, T)\}, \quad R_{L_d}(\Gamma, T) = (I - L_d^{-1} \Gamma T)^{L_d}. \quad (7.1)$$

This identity is well defined whenever the displayed product is trace class. It is the RKHS lift of the finite-dimensional residual, not an identification of softmax weights with T . The prescribed nonlinear attention determines T or its normalized analogue; the recurrent calculation begins after that operator has been fixed.

7.1 Isotropic and fixed-spectrum tasks

If $T = \lambda I$ and $\Omega = \omega I$ on a finite represented space, orthogonal invariance and zero initialization keep $\Gamma(t) = \gamma(t)I$. For a fixed structured operator, assume that T and Ω commute, with eigenpairs (λ_j, ω_j) . The eigenvalues γ_j of Γ then decouple.

Proposition 7.1 (Kernel-isotropic finite-context reduction). *Let the symmetrized empirical kernel operator $\hat{T}_{\kappa, n}$ have an orthogonally invariant law and limiting spectral measure $\rho_{\kappa, \alpha}$, where α denotes the context-to-dimension ratio in a finite-dimensional approximation. For isotropic task covariance and zero initialization, a global reduced head remains $\Gamma(t) = \gamma(t)I$, and*

$$\dot{\gamma}(t) = \int \lambda \left(1 - L_d^{-1} \gamma(t) \lambda\right)^{2L_d-1} \rho_{\kappa, \alpha}(d\lambda), \quad (7.2)$$

$$\mathcal{L}_\star(\alpha, L_d) = \inf_{\gamma} \int \left(1 - L_d^{-1} \gamma \lambda\right)^{2L_d} \rho_{\kappa, \alpha}(d\lambda). \quad (7.3)$$

Thus the Marchenko–Pastur law of isotropic linear features is replaced by the spectral law of the chosen normalized kernel. No other change to the reduced calculus is valid without deriving that law.

Proposition 7.2 (Fixed-spectrum gradient flow). *For gradient flow $\dot{\Gamma} = -\nabla_{\Gamma} \mathcal{L}_{L_d}/2$ initialized at zero, the commuting subspace is invariant and, up to the inessential global normalization of time,*

$$\dot{\gamma}_j(t) = \omega_j \lambda_j^2 \left(1 - L_d^{-1} \lambda_j \gamma_j(t)\right)^{2L_d-1}. \quad (7.4)$$

On every finite represented subspace, the zero-loss fixed point is $\Gamma = L_d T^{-1}$; hence one effective step is sufficient after the covariance has been memorized. In the large-depth limit, choosing the time normalization with a factor two gives

$$\gamma_j(t) = \frac{1}{2\lambda_j} \log(1 + 4\omega_j \lambda_j^3 t), \quad \mathcal{L}_{\infty}(t) = \sum_j \frac{\omega_j \lambda_j}{1 + 4\omega_j \lambda_j^3 t}. \quad (7.5)$$

In infinite dimension, T^{-1} is generally unbounded. The statement that a fixed covariance can be inverted in one layer therefore needs either finite width or regularization. For GP/KRR regression, replacing T by $T_{\tau} = T + \tau I$, $\tau > 0$, yields the bounded fixed point $L_d T_{\tau}^{-1}$. This is the operator-theoretic reason that posterior whitening is safer than memorizing an unregularized covariance inverse.

The following formulation avoids ambiguity in source/capacity conventions.

Corollary 7.3 (Regular-variation form of the fixed-spectrum time law). *Suppose*

$$a_j := \omega_j \lambda_j \sim c_a j^{-s}, \quad b_j := \omega_j \lambda_j^3 \sim c_b j^{-r}, \quad r > s - 1 > 0. \quad (7.6)$$

Then

$$\mathcal{L}_{\infty}(t) = \Theta\left(t^{-(s-1)/r}\right). \quad (7.7)$$

For the convention $\lambda_j \asymp j^{-\nu}$ and $\sum_{\ell > j} \omega_{\ell} \lambda_{\ell} \asymp j^{-\nu\beta}$, one has $s = \nu\beta + 1$ and $r = s + 2\nu$, so the exponent follows directly from (7.7). Stating the law through (s, r) prevents changes of capacity convention from silently changing the exponent.

If a fully trained fixed-spectrum head is tested on T' while retaining $\Gamma = L_d T^{-1}$, the exact out-of-distribution residual is

$$R_{\text{OOD}} = (I - T^{-1} T')^{L_d}, \quad \mathcal{L}_{\text{OOD}} = \text{tr}\{\Omega' R_{\text{OOD}}^* T' R_{\text{OOD}}\}. \quad (7.8)$$

Thus the brittleness of a globally stored whitening map survives unchanged in RKHS. Prompt-conditioned posterior geometry is designed precisely to avoid using the same inverse eigenspace for every task.

7.2 Randomly rotated spectra

Let $T_U = U T U^*$ with an independent Haar rotation in each finite-dimensional approximation. By the same invariance behind Proposition 9.1, population gradient flow from zero keeps a global Γ isotropic. Writing $\Gamma = \gamma I$ gives

$$\mathcal{L}_{L_d}(\gamma) = \sum_j \omega_j \lambda_j \left(1 - \frac{\gamma \lambda_j}{L_d}\right)^{2L_d}, \quad \dot{\gamma}(t) = \sum_j \omega_j \lambda_j^2 \left(1 - \frac{\gamma \lambda_j}{L_d}\right)^{2L_d-1}. \quad (7.9)$$

Proposition 7.4 (Source/capacity laws for the rotated model). *Assume*

$$\lambda_j \asymp j^{-\nu}, \quad \sum_{k > j} \omega_k \lambda_k \asymp j^{-\nu\beta}, \quad \nu, \beta > 0. \quad (7.10)$$

Then, in the infinite-depth population model,

$$\mathcal{L}_\infty(\gamma) \asymp \gamma^{-\beta}, \quad \gamma(t) \asymp t^{1/(\beta+2)}, \quad \mathcal{L}_\infty(t) \asymp t^{-\beta/(\beta+2)}. \quad (7.11)$$

At infinite training time, stability imposes $\gamma = O(L_d/\lambda_1)$ and

$$\inf_{\gamma} \mathcal{L}_{L_d}(\gamma) = \Theta(L_d^{-\beta}). \quad (7.12)$$

A spectral rank- S feature bottleneck or an idealized noiseless context that reveals the first n modes produces, respectively, the marginal laws $S^{-\nu\beta}$ and $n^{-\nu\beta}$. A random-design context requires the corresponding kernel random-matrix argument.

The noisy Bayesian context law (8.7) is different from the noiseless rank law in Proposition 7.4. The former sums posterior variances; the latter counts inaccessible target energy. Equating them would conflate observation noise with a rank bottleneck.

7.3 Parameterization and compute-optimal shape

The outer kernel nonlinearity is fixed, but the trainable amplitude can still be parameterized in different ways. This changes optimization time without changing the represented kernel.

Proposition 7.5 (Balanced amplitude parameterization). *Suppose the reduced scale is parameterized as $\gamma = w^r$, $r \geq 1$, and w follows gradient flow. If $\mathcal{L}(\gamma) \asymp \gamma^{-\beta}$, then*

$$\gamma(t) \asymp t^{1/(\beta+2/r)}, \quad \mathcal{L}(t) \asymp t^{-r\beta/(r\beta+2)}. \quad (7.13)$$

The reduced- Γ model has $r = 1$. The balanced five-factor linear attention parameterization has $r = 5$ and therefore exponent $5\beta/(5\beta + 2)$.

Combining the four marginal rotated-spectrum laws gives the operator analogue of the separable scaling envelope

$$\mathcal{L}(t, S, L_d, n) \approx c_t t^{-r\beta/(r\beta+2)} + c_S S^{-\nu\beta} + c_L L_d^{-\beta} + c_n n^{-\nu\beta}. \quad (7.14)$$

This is a marginal asymptotic statement: the exact joint interactions remain inside the spectral formula. Under the attention-compute model $\mathfrak{C} \asymp t n^2 S^2 L_d$, balancing the width and unpreconditioned depth terms gives

$$L_d \asymp S^\nu. \quad (7.15)$$

Our posterior metric changes this conclusion when Theorem 4.2 provides a uniform effective condition number. The algebraic term $L_d^{-\beta}$ is then replaced by $[1 + L_d(1 + q)]^2 q^{2L_d}$ for HB or $4q^{2L_d}$ for Chebyshev. Balancing it against $S^{-\nu\beta}$ gives, to leading order,

$$L_d \asymp \frac{\nu\beta}{2 \log(1/q)} \log S \quad (7.16)$$

up to $\log \log S$ for the Heavy-Ball transient. Thus the central predicted architectural change is from polynomial depth-width scaling to logarithmic depth once a mesh-uniform posterior metric has been learned.

7.4 Two-point resolvents and nonlinear kernels

Finite width and finite context need not commute with a random task rotation. Let $M_{\kappa,h}$ denote the symmetrized empirical kernel operator composed with the rank- S feature projection. For any polynomial p and contours $\mathcal{C}, \mathcal{C}'$ enclosing its spectrum, Cauchy's formula gives the exact identity

$$\begin{aligned} & \text{tr}\{\Omega p(M_{\kappa,h})^* T p(M_{\kappa,h})\} \\ &= \frac{1}{(2\pi i)^2} \oint_{\mathcal{C}} \oint_{\mathcal{C}'} p(z) p(w) \underbrace{\text{tr}\{\Omega(zI - M_{\kappa,h})^{-1} T(wI - M_{\kappa,h})^{-1}\}}_{C_{\kappa,h}(z,w)} dz dw. \end{aligned} \quad (7.17)$$

This is the nonlinear-kernel version of the two-point reduction: every finite-depth risk depends on the two-resolvent correlation $C_{\kappa,h}$, not only on the eigenvalue density. A random-matrix or DMFT limit must derive $C_{\kappa,h} \rightarrow C_\kappa$ for the chosen kernel and sampling law. The Wishart/free-product deterministic equivalent of linear attention cannot simply be reused for row-normalized softmax. Once C_κ is known, however, the contour integral and the HB/Chebyshev substitution are exact.

If the layer scales $\gamma_1, \dots, \gamma_{L_d}$ are untied but the population loss is permutation symmetric and they start equally, gradient flow preserves $\gamma_1 = \dots = \gamma_{L_d}$. Hence the tied reduced model describes that balanced trajectory. This symmetry statement is independent of whether T came from linear, RBF-softmax, or another prescribed kernel.

Finite-batch boundary. The shallow online-SGD calculation in the linear isotropic model closes using Gaussian fourth moments. For kernel attention the exact one-step identity is still

$$\mathbb{E}_t \|\Gamma_{t+1} - \Gamma_\star\|_{\text{HS}}^2 = \|\Gamma_t - \Gamma_\star - \eta \nabla \mathcal{L}(\Gamma_t)\|_{\text{HS}}^2 + \eta^2 \mathbb{E}_t \|\xi_{\kappa,t}\|_{\text{HS}}^2, \quad (7.18)$$

where $\xi_{\kappa,t}$ is the centered mini-batch gradient noise. Closing this recursion requires kernel-specific four-point correlations, not merely $\rho_{\kappa,\alpha}$. We therefore retain the exact identity but do not import the linear-Wishart coefficients into nonlinear softmax attention.

7.5 Transfer map from the linear scaling theory

Table 1 records the status of each structural calculation used in the linear theory. “Exact” means exact under the assumptions stated here, rather than an analogy suggested by matching exponents.

Linear-attention object	Kernel-matched replacement	Status
ISO scalar gradient flow	Empirical normalized-kernel spectral law $\rho_{\kappa,\alpha}$	Proposition 7.1
Fixed-spectrum whitening	Bounded KRR/posterior inverse on represented modes	Proposition 7.2
Fixed-spectrum OOD loss	Operator residual $(I - T^{-1}T')^{L_d}$	Equation (7.8)
Random rotations	Isotropic global metric; contextual Ritz subspace	Propositions 7.4 and 9.1
Two-point deterministic equivalent	Kernel two-resolvent correlation $C_{\kappa,h}$	Exact reduction; kernel limit required
Width, depth, context, time	Equations (7.14) and (8.5)	Exact in their stated modal models
Full weight parameterization	General balanced map $\gamma = w^r$	Proposition 7.5
Untied layers	Balanced permutation-symmetric trajectory	Exact symmetry reduction
Softmax nonlinearity	Prescribed normalized kernel plus quadrature	Theorem 3.5
Finite-batch SGD coefficients	Kernel four-point gradient-noise law	Equation (7.18); not universal

Table 1: What transfers from the reduced linear-attention calculations.

8 A joint scaling law in the commuting GP model

We next give an exactly solvable model which isolates the five resources. It is the infinite-dimensional analogue of simultaneously diagonalizable fixed-spectrum regression. Let

$$f = \sum_{j \geq 1} f_j e_j, \quad f_j \stackrel{\text{ind}}{\sim} \mathcal{N}(0, \lambda_j), \quad Y_j = f_j + \frac{\sigma}{\sqrt{n}} \xi_j, \quad \xi_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (8.1)$$

Here n is context or information size. The sequence model is an exact white-noise experiment; for random-design kernel regression, transferring its formula requires an asymptotic-equivalence or concentration argument and is not assumed silently.

The posterior variance, posterior-mean variance, and shrinkage are

$$r_{n,j} = \frac{\sigma^2 \lambda_j}{\sigma^2 + n \lambda_j}, \quad v_{n,j} = \lambda_j - r_{n,j} = \frac{n \lambda_j^2}{\sigma^2 + n \lambda_j}, \quad s_{n,j} = \frac{n \lambda_j}{\sigma^2 + n \lambda_j}. \quad (8.2)$$

Effective width S means that only the first S kernel modes are represented. A discretization h represents J_h modes. To describe training time, let $\chi_j(t) \in [0, 1]$ be the fraction of the j th posterior-mean mode acquired by the feature head after time t ; modal gradient flow gives

$$\chi_j(t) = 1 - e^{-\gamma_j t}. \quad (8.3)$$

Finally, let $p_L(\zeta_{h,j})$ be the residual of the exact depth- L loop on the effective preconditioned eigenvalue $\zeta_{h,j}$. Set $M_h = \min(S, J_h)$.

Theorem 8.1 (Exact depth-width-context-time risk). *In the commuting model, if the approximate conditional mean is*

$$\hat{f}_{L,j} = \begin{cases} [1 - p_L(\zeta_{h,j})] \chi_j(t) \mathbb{E}[f_j | Y_j], & j \leq M_h, \\ 0, & j > M_h, \end{cases} \quad (8.4)$$

then its integrated Bayes risk is exactly

$$\mathcal{R}(n, S, L, t, h) = \sum_{j \leq M_h} \left\{ r_{n,j} + [1 - (1 - p_L(\zeta_{h,j})) \chi_j(t)]^2 v_{n,j} \right\} + \sum_{j > M_h} \lambda_j. \quad (8.5)$$

If the spatial discretization additionally perturbs represented modes by an order- p consistent method, the right-hand side changes by at most Ch^{2p} in squared prediction risk under the corresponding stability assumption.

Formula (8.5) is more informative than multiplying separate power laws. Context fixes the irreducible conditional variance $r_{n,j}$; width and discretization decide which modes exist; training determines which of those modes have been acquired; and depth filters the acquired posterior mean by an exact solver polynomial.

8.1 Power-law consequences

Assumption 8.2 (Capacity and optimization spectra). *For constants $a > 1$, $b > 0$, $c > 1$, and positive c_λ, c_γ, c_w ,*

$$\lambda_j \sim c_\lambda j^{-a}, \quad \gamma_j \sim c_\gamma j^{-b}, \quad w_j \sim c_w j^{-c}. \quad (8.6)$$

The weights w_j specify the risk carried by an unlearned mode. In the well-specified GP Bayes risk one has $w_j = \lambda_j$ asymptotically.

Theorem 8.3 (Marginal scaling laws). *Under Assumption 8.2:*

$$\sum_{j \geq 1} r_{n,j} \sim c_\lambda^{1/a} \frac{\pi}{a \sin(\pi/a)} \left(\frac{\sigma^2}{n} \right)^{1-1/a}, \quad (8.7)$$

$$\sum_{j > S} \lambda_j \sim \frac{c_\lambda}{a-1} S^{1-a}, \quad (8.8)$$

$$\sum_{j \geq 1} w_j e^{-2\gamma_j t} \sim \frac{c_w}{b} \Gamma\left(\frac{c-1}{b}\right) (2c_\gamma t)^{-(c-1)/b}. \quad (8.9)$$

If the mesh-independent effective spectrum is contained in $[\mu, L]$, the depth term is bounded by

$$\mathcal{E}_{\text{depth}}^{\text{HB}} \leq C[1 + L_d(1 + q)]^2 q^{2L_d}, \quad \mathcal{E}_{\text{depth}}^{\text{Ch}} \leq 4Cq^{2L_d}, \quad (8.10)$$

where L_d denotes loop depth and $q = (\sqrt{L/\mu} - 1)/(\sqrt{L/\mu} + 1)$.

The two uses of L in the literature—spectral upper endpoint and depth—are separated as L and L_d in (8.10). Combining the bounds gives the transparent envelope

$$\mathcal{R} - \mathcal{R}_\star \lesssim n^{-(a-1)/a} + S^{1-a} + t^{-(c-1)/b} + \mathcal{E}_{\text{depth}}(L_d) + h^{2p}, \quad (8.11)$$

with the exact interactions retained in (8.5). The slowest term is the bottleneck. In particular, extra depth cannot repair a missing kernel mode, and extra width cannot repair a posterior metric learned in the wrong feature space.

9 Fixed and rotating posterior spectra

The finite-dimensional fixed-spectrum/randomly-rotated-spectrum distinction extends directly to whitened posterior operators. Let

$$H_U = I + U \text{diag}(\nu_1, \dots, \nu_d) U^\top. \quad (9.1)$$

In a fixed-spectrum family, U is shared and a global head can store its inverse geometry. In a rotated family, U changes independently between prompts.

Proposition 9.1 (A global metric cannot identify random eigenspaces). *Let U be Haar distributed on the orthogonal group and independent across tasks. Among deterministic matrices B , the unique minimizer of*

$$\mathbb{E}_U \|B - H_U^{-1}\|_F^2 \quad (9.2)$$

is

$$B_\star = \mathbb{E}_U H_U^{-1} = \left[\frac{1}{d} \sum_{j=1}^d \frac{1}{1 + \nu_j} \right] I. \quad (9.3)$$

Thus a prompt-independent head can learn only an isotropic average. A prompt-conditioned head can outperform it only by extracting information about U from the context.

Nonlinear attention does not evade this symmetry by itself. Its value is that kernel-matched contextual fields can identify the task-specific posterior subspace before the exact Ritz and polynomial operations are applied. This is the function-space analogue of moving from a stored global preconditioner to an in-context algorithm.

10 Inverse PDE specialization

Let m be a coefficient field and let $u(m; f)$ solve a PDE. Weak observations of pairs (u_i, f_i) define an observation map G in the linear case or a Jacobian $J(m_k)$ in a local nonlinear model. The functions in one prompt may be sampled on different point clouds. Their coordinates, quadrature weights, field labels, and boundary masks are tokens. Kernel-matched attention maps these heterogeneous observations to continuum feature fields; it does not first interpolate every function to a privileged uniform grid.

For a linear inverse problem, the whitened system is exactly (2.2) [11]. For a nonlinear forward map $\mathcal{G}$, a Gauss–Newton–Laplace step uses

$$H_k = I + C_0^{1/2} J(m_k)^* \Gamma_\varepsilon^{-1} J(m_k) C_0^{1/2}, \quad H_k \delta_k = g_k. \quad (10.1)$$

The attention head is evaluated on the current prompt and linearization, then held fixed during the inner HB, Chebyshev, or PCG solve.

Proposition 10.1 (Local nonlinear extension). *Assume that $\mathcal{G}$ is continuously Fréchet differentiable in a neighbourhood of a locally identifiable solution, its Jacobian is Lipschitz, the regularized Gauss–Newton Hessians are uniformly coercive and bounded, and the inner residuals satisfy the standard forcing condition*

$$\|H_k \delta_k - g_k\| \leq \eta_k \|g_k\|, \quad \sup_k \eta_k < 1. \quad (10.2)$$

If the learned metrics are uniformly spectrally equivalent to H_k^{-1} , then the number of polynomial inner iterations needed to enforce (10.2) is uniform in the mesh. The usual local inexact Gauss–Newton convergence conclusion then holds; if $\eta_k \rightarrow 0$, the inner error does not obstruct superlinear convergence under the corresponding standard assumptions.

This is a local Gaussian/Laplace statement. Sampling a multimodal posterior would require a different outer method, such as function-space MCMC or a diffusion model, and is outside the scope of this paper.

11 Numerical validation and claim boundaries

We validate the deterministic identities before testing any scaling or timing claim. All nonlinear-kernel experiments use the prescribed RBF score and quadrature weights; its length scale is fixed by the covariance model and is never optimized from the reported test data. Raw samples, configuration, and failure-preserving summaries accompany this manuscript.

Algebra, RMT, and reduced DMFT. The weighted softmax/kernel identity has maximum relative residual 7.859×10^{-16} , and no nonvacuous Ritz spectral certificate is violated. Monte Carlo fixed polynomial risk agrees with the exact trace formula to relative error 0.007222. The Marchenko–Pastur comparison is used only as a linear-Wishart control. For the normalized RBF operator we instead measure its own one- and two-resolvent statistics; their penultimate/coarsest finite-size discrepancy ratio is 0.1494. Thus these experiments support convergence of the probed kernel statistics, not an unproved closed-form nonlinear deterministic equivalent. The FS/RRS time, depth, width/context, balanced-parameterization, and finite-task isotropy slopes pass their prespecified finite-window tolerance and R^2 criteria. Regression standard errors are reported diagnostically; deterministic modal-truncation bias is not treated as sampling noise.

Table 2: Measured log–log exponents against the reduced-theory values.

Law	measured	theory	R^2
FS time	-0.3158	-0.3158	1
RRS loss versus scale	-1.506	-1.5	1
RRS time, $r = 1$	-0.4302	-0.4286	1
RRS time, $r = 5$	-0.7928	-0.7895	1
RRS width/context	-1.198	-1.2	1
RRS depth	-1.558	-1.5	1
Finite-task isotropy	-0.4967	-0.5	0.9999

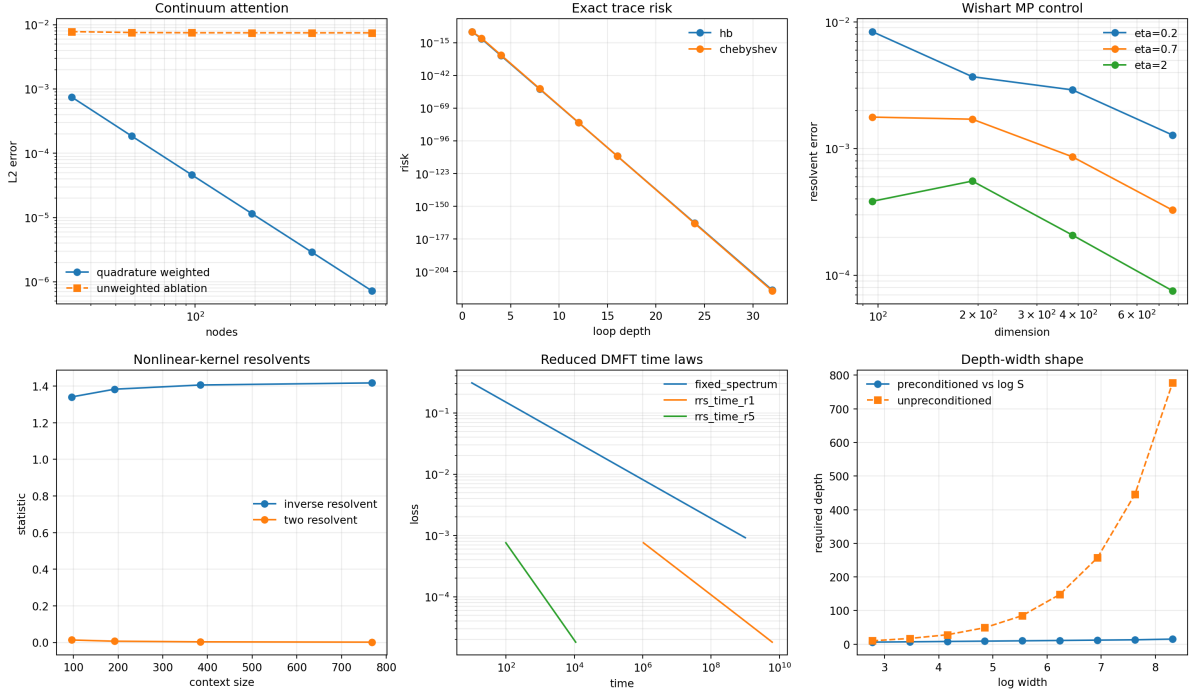


Figure 1: Algebraic, RMT, nonlinear-resolvent, reduced-DMFT, and depth-width validations. Marchenko–Pastur is shown only for the linear control.

Discretization covariance. On nonuniform meshes the lifted Ritz metric converges with fitted order 2 and the transfer commutator with order 2.437. Removing quadrature weights increases the finest-grid metric error by a factor 4.037×10^4 . This directly separates continuum covariance from mere variable-length execution.

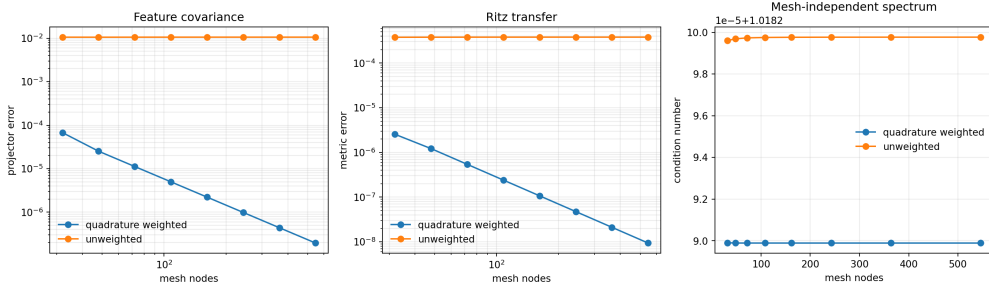


Figure 2: Nonuniform-mesh transfer. Quadrature weighting converges to one lifted projector and Ritz metric, while the unweighted head retains a sampling-measure bias.

Elliptic Bayesian inverse problem. We solve $-\nabla \cdot (a_z \nabla u) + 0.2u = m$ on the unit square with homogeneous Dirichlet data. The log-diffusion is a bounded random Fourier field and the source covariance square root is $W_z = \sigma_f I + \Phi_z \text{diag}(d_z) \Phi_z^\top$, with a task-dependent localized rank-24 component and a nonzero floor. Point observations and sparse elliptic solves produce the prior-whitened posterior $H_z = I + U_z U_z^\top$. The model-chosen RBF length scale is compared with short- and long-scale ablations before one exact block-power refinement and the Ritz construction. The largest standard sweep has 16384 state unknowns and 512 observation tokens. The contextual kernel–Ritz metric reduces the effective condition number by a factor 885.7 and the fixed four-HVP loops meet the common residual tolerance. At this moderate context, exact Woodbury is faster; its one-query total divided by kernel–HB is 0.2137. Accordingly we make no universal speed claim against a structure-exploiting Woodbury solve.

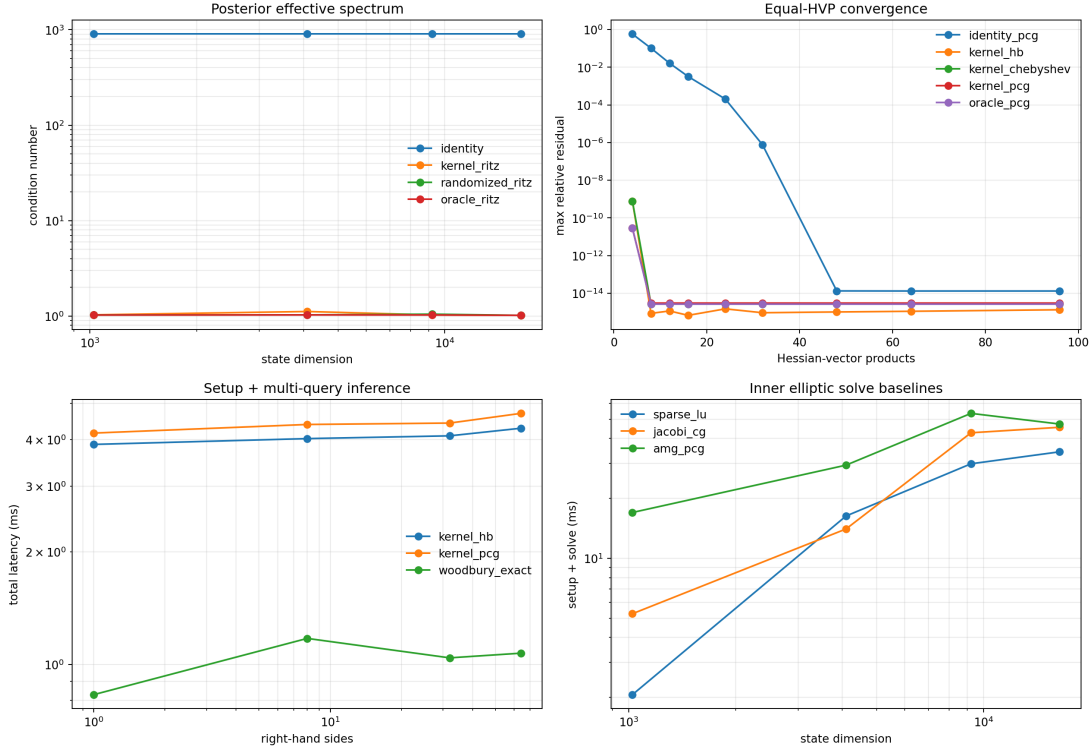


Figure 3: Variable-coefficient elliptic posterior validation: effective spectra, equal-HVP convergence, multi-query timing, and sparse LU/Jacobi/AMG inner PDE baselines.

Table 3: Long-context setup-plus-one-query latency on the same posterior system on an NVIDIA H100 PCIe. Parentheses give bootstrap 95% intervals from 11 repetitions.

Method	depth	latency (ms)	maximum relative residual
Kernel-Ritz HB	10	10.9 [10.87,11.27]	2.268×10^{-8}
Exact Woodbury	—	24.03 [23.38,24.26]	1.541×10^{-15}
Dense Cholesky	—	139.8 [135.5,141.3]	1.849×10^{-14}

At state dimension $N = 16384$ and context length $m = 4096$, the measured dense/kernel median ratio is 12.82. A nonoverlapping Woodbury crossover is also observed. The admissible inference-speed statement is therefore restricted to the measured context/query region and always includes feature construction. PDE assembly is a common context cost and is reported separately; AMG-PCG and sparse LU are included as inner elliptic baselines.

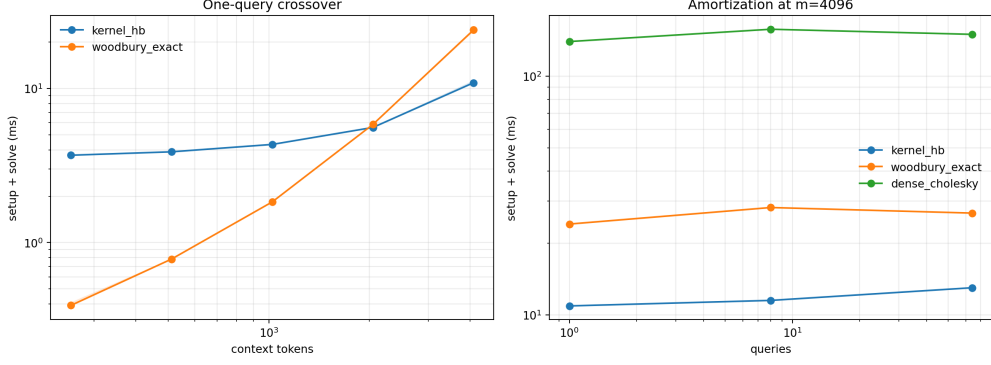


Figure 4: Setup-inclusive long-context crossover and multi-query amortization. Exact Woodbury is faster at short context; the compressed kernel–Ritz loop crosses it only once the context is sufficiently long.

12 Discussion and limitations

The modelling choice. The kernel family and its geometry-defining hyperparameters are fixed. This matches a Bayesian analysis in which the prior has been specified before inference. Marginal-likelihood tuning, mixtures of kernels, or learned input warps are valuable but distinct empirical-Bayes models; allowing them would change the feature space during training and invalidate the clean resource separation in Theorem 8.1.

What is and is not learned. The encoder learns prompt-dependent coefficients and a finite subspace within the prescribed feature geometry. It does not learn the Hessian–vector product, orthonormalization, projected inverse, Heavy–Ball arithmetic, or Chebyshev recurrence. This makes positivity and solver stability properties of the architecture rather than regularization preferences.

Comparison to classical solvers. Finite elements, operator preconditioning, domain decomposition, and AMG can already be mesh robust. The proposed novelty is not irregular-mesh competence. It is amortized inference of a posterior metric from function pairs rather than construction from a supplied system matrix, combined with a discretization-covariance theorem and an exact loop. Section 11 shows that runtime advantage only in a bounded regime: for a long context and one or a few queries, the rank- S contextual metric avoids both an $N \times N$ Cholesky factorization and the $m \times m$ Woodbury setup. At moderate context exact Woodbury is faster, and with enough repeated queries its setup can be amortized. We therefore do not claim universal domination of classical structure-exploiting solvers.

Limits of the scaling law. Equation (8.5) is exact in the commuting Gaussian sequence model. Applying the same exponents to random-design, noncommuting, or fully feature-learning Transformers needs additional concentration, random-matrix, or DMFT analysis. The deterministic trace formula (6.8) states precisely what such an analysis must supply: the weighted spectral law of the prompt-conditioned preconditioned Hessian.

13 Conclusion

The nonlinearity of attention should reflect the covariance geometry assumed by the modeller. Exponential softmax is the correct normalized interaction for an RBF-type kernel with fixed scale; it is not a universal representation of Gaussian-process covariance. Once quadrature and normalization are made explicit, kernel attention becomes a consistent continuum operation. Its contextual outputs can be converted by exact Ritz algebra into an SPD approximation of posterior covariance, after which tied Heavy-Ball or Chebyshev layers perform auditable inference. The resulting theory separates five resources—context, width, training time, depth, and discretization—and shows how mesh-independent posterior geometry yields mesh-independent looped inference. The numerical audit supports the exact spectral and reduced-model predictions, rejects importing the linear Marchenko–Pastur law into nonlinear softmax, and locates a setup-inclusive long-context crossover against both dense Cholesky and exact Woodbury. This measured boundary is part of the claim: outside it, the appropriate classical solver should be used.

A Proofs for continuum kernel attention

Proof of Proposition 3.3. For $f, g \in L^2(\pi_\kappa)$, Fubini's theorem and symmetry give

$$\begin{aligned}\langle f, \mathcal{A}_\kappa g \rangle_{L^2(\pi_\kappa)} &= \frac{1}{Z_\kappa} \int_X \int_X f(x) \kappa(x, y) g(y) \rho(dy) \rho(dx) \\ &= \langle \mathcal{A}_\kappa f, g \rangle_{L^2(\pi_\kappa)}.\end{aligned}$$

Taking $f = g$ shows nonnegativity whenever κ is a positive-semidefinite integral kernel. Discretely,

$$\pi_{h,i} a_{ij}^{\kappa,h} \propto \omega_i \omega_j \kappa(x_i, x_j) = \pi_{h,j} a_{ji}^{\kappa,h},$$

which is detailed balance. The same quadratic-form calculation proves positivity. $\square$

Proof of Theorem 3.5. Write $N(x) = \int \kappa(x, y) v(y) \rho(dy)$ and $D(x) = d_\kappa(x)$, with quadrature approximations N_h, D_h . Assumption 3.4 gives $|N_h - N| \leq C_v h^p$ and $|D_h - D| \leq C_0 h^p$. For sufficiently small h , $D_h \geq d_0/2$. Hence

$$\left| \frac{N_h}{D_h} - \frac{N}{D} \right| \leq \frac{|N_h - N|}{|D_h|} + \frac{|N| |D_h - D|}{|D_h| D} \leq C(1 + \|v\|_\infty) h^p.$$

The finitely many coarser admissible discretizations are absorbed by enlarging C . $\square$

B Proofs for the posterior metric

Proof of Proposition 4.1. Every $x \in \mathcal{U}$ decomposes as $x = Qa + x_\perp$ with $Q^* x_\perp = 0$. Since $H \succeq I$,

$$\langle x, B_Q x \rangle = \|x_\perp\|^2 + \langle a, (Q^* H Q)^{-1} a \rangle > 0$$

for $x \neq 0$, and $(Q^* H Q)^{-1} \preceq I$ proves $B_Q \preceq I$. If $\text{range}(A) \subseteq \text{range}(Q)$, both $\text{range}(Q)$ and its orthogonal complement are invariant under H . On the complement $H = I$, while on $\text{range}(Q)$ the second term in (4.1) is the exact inverse of the restriction. Thus $B_Q = H^{-1}$. Rearranging (4.1) gives (4.2); the bracket is positive semidefinite because $Q^* H Q \succeq I$. $\square$

Lemma B.1 (Ritz map perturbation). *Let $I \preceq H_1, H_2 \preceq MI$ and let $P_i = Q_i Q_i^*$ be rank- S orthogonal projectors. After optimal right-orthogonal alignment of Q_1, Q_2 ,*

$$\|B(H_1, Q_1) - B(H_2, Q_2)\|_{\text{op}} \leq \|H_1 - H_2\|_{\text{op}} + C_M \|P_1 - P_2\|_{\text{op}}, \quad (\text{B.1})$$

where $B(H, Q)$ denotes (4.1) and C_M may be chosen as in (4.3) after increasing it by an absolute constant.

Proof. The principal-angle relation gives an alignment with $\|Q_1 - Q_2\|_{\text{op}} \leq \sqrt{2} \|P_1 - P_2\|_{\text{op}}$. Add and subtract the two complementary projectors and the two Ritz terms. The resolvent identity

$$R_1^{-1} - R_2^{-1} = R_1^{-1} (R_2 - R_1) R_2^{-1}, \quad R_i = Q_i^* H_i Q_i,$$

has $\|R_i^{-1}\| \leq 1$. Moreover

$$\|R_1 - R_2\| \leq \|H_1 - H_2\| + 2M \|Q_1 - Q_2\|.$$

The outer factors contribute another $2\|Q_1 - Q_2\|$, and the complementary projectors contribute $\|P_1 - P_2\|$. Combining the bounds proves (B.1) with the advertised nonoptimal constant. $\square$

Proof of Theorem 4.2. Let $Q_\star$ span the first S eigenvectors. The spectral theorem gives

$$\|B_{Q_\star} - H^{-1}\|_{\text{op}} = \sup_{j>S} \left| 1 - \frac{1}{1 + \nu_j} \right| = r_S.$$

Lemma B.1 with $H_1 = H_2 = H$ gives $\|B_Q - B_{Q_\star}\| \leq C_M \delta_S$. This proves (4.4). Consequently

$$\|H^{1/2} B_Q H^{1/2} - I\|_{\text{op}} \leq M(r_S + C_M \delta_S) = \epsilon_S.$$

Weyl's inequality places the spectrum of $H^{1/2} B_Q H^{1/2}$ in the stated interval. It has the same spectrum as $B_Q^{1/2} H B_Q^{1/2}$, which proves (4.6). $\square$

Proof of Theorem 5.2. Apply Lemma B.1 to $(\bar{H}_h, \bar{P}_h)$ and (H, P) , then use (5.2); this gives (5.3). Insert and subtract the continuum B between the two lifted discrete metrics. The triangle inequality gives the sum of the two one-level errors. Compressing back to the two discrete spaces yields (5.4); any failure of the transfer to equal exact lifted compression is, by definition, ε_{tr} . $\square$

Proof of Corollary 5.3. Theorem 5.2 and (5.2) bound the lifted symmetrically preconditioned perturbation by a term tending to an encoder-dependent floor. By hypothesis that floor is below $\mu_0/2$, and h can be chosen so that the full perturbation remains below $\mu_0/2$. Spectra of bounded self-adjoint operators are stable under operator-norm perturbations; one may then take $\mu = \mu_0/2$ and enlarge L_0 by the same uniform perturbation bound. $\square$

C Proofs for looped inference

Proof of Theorem 6.1. Let $e_\ell = x_\ell - x_\star$ and $\tilde{e}_\ell = B^{-1/2} e_\ell$. Then

$$\tilde{e}_{\ell+1} = [(1 + \beta)I - \alpha K] \tilde{e}_\ell - \beta \tilde{e}_{\ell-1},$$

so $\tilde{e}_\ell = p_\ell(K) \tilde{e}_0$. For the coefficients (6.4), write $1 + \beta_\star - \alpha_\star z = 2q\tau$, where $\tau \in [-1, 1]$ as z ranges over $[\mu, L]$. Setting $s_\ell = p_\ell/q^\ell$ gives $s_{\ell+1} = 2\tau s_\ell - s_{\ell-1}$ and hence

$$p_\ell(z) = q^\ell [U_\ell(\tau) - q U_{\ell-1}(\tau)].$$

Since $|U_\ell(\tau)| \leq \ell + 1$ on $[-1, 1]$,

$$\max_{z \in [\mu, L]} |p_\ell(z)| \leq [1 + \ell(1 + q)] q^\ell.$$

Functional calculus proves (6.5). The strict stability region (6.6) follows from the Jury conditions for $r^2 - (1 + \beta - \alpha z)r + \beta$, uniformly over $z \in [\mu, L]$. $\square$

Proof of Proposition 6.2. In the variable $u = B^{-1/2}x$, the exact solution is $u_\star = K^{-1}c$ and a fixed polynomial method has error $p_\ell(K)u_\star$. Therefore

$$\|x_\ell - x_\star\|_H^2 = \langle p_\ell(K)K^{-1}c, K p_\ell(K)K^{-1}c \rangle = \langle c, K^{-1}p_\ell(K)^2 c \rangle.$$

Taking the conditional expectation gives the trace. The final expression is the spectral theorem applied to the positive weighted measure $\nu_c(E) = \text{tr}(\Sigma_c \mathbf{1}_E(K))$. $\square$

D Proofs for the reduced preconditioner operator calculus

Proof of Proposition 7.1. Conjugation invariance of the population loss implies that its gradient at a scalar operator is scalar. Zero initialization therefore leaves the scalar manifold invariant. Conditional on $\widehat{T}_{\kappa,n}$, spectral calculus turns the normalized trace loss into the empirical average of $(1 - L_d^{-1}\gamma\lambda)^{2L_d}$. Differentiating gives the empirical version of (7.2). Convergence of the empirical spectral measure, together with uniform integrability on the stable parameter range, gives (7.2)–(7.3). $\square$

Proof of Proposition 7.2. Simultaneous diagonalization turns (7.1) into

$$\mathcal{L}_{L_d} = \sum_j \omega_j \lambda_j \left(1 - L_d^{-1} \lambda_j \gamma_j\right)^{2L_d}.$$

Differentiation gives (7.4), up to the stated constant rescaling of time. The zero of each represented residual is $\gamma_j = L_d/\lambda_j$. As $L_d \rightarrow \infty$, $(1 - L_d^{-1} \lambda_j \gamma_j)^{2L_d-1} \rightarrow e^{-2\lambda_j \gamma_j}$. With the factor-two convention,

$$\dot{\gamma}_j = 2\omega_j \lambda_j^2 e^{-2\lambda_j \gamma_j}.$$

Multiplying by $e^{2\lambda_j \gamma_j}$ and integrating from zero gives $e^{2\lambda_j \gamma_j(t)} = 1 + 4\omega_j \lambda_j^3 t$. Substitution in the modal loss proves (7.5). $\square$

Proof of Corollary 7.3. Integral comparison for regularly varying sequences reduces the asymptotic, up to positive constants, to

$$\int_1^\infty \frac{x^{-s}}{1+tx^{-r}} dx.$$

The substitution $x = t^{1/r}u$ gives $t^{-(s-1)/r}$ times an integral converging at both zero and infinity exactly when $r > s-1 > 0$. This proves (7.7). Under the displayed source condition, Karamata's theorem gives $a_j \asymp j^{-\nu\beta-1}$; multiplying by λ_j^2 gives $b_j \asymp j^{-(\nu\beta+1+2\nu)}$. $\square$

Proof of Proposition 7.4. At infinite depth,

$$\mathcal{L}_\infty(\gamma) = \sum_j a_j e^{-2\gamma\lambda_j}, \quad a_j = \omega_j \lambda_j.$$

Let $J_\gamma = \lfloor \gamma^{1/\nu} \rfloor$. Modes $j \gg J_\gamma$ have $\gamma\lambda_j = O(1)$ and contribute a constant fraction of their tail energy, which is $\Theta(J_\gamma^{-\nu\beta})$. Modes $j \ll J_\gamma$ are exponentially suppressed; a dyadic decomposition around J_γ bounds their total by the same order. Hence $\mathcal{L}_\infty(\gamma) = \Theta(\gamma^{-\beta})$. Differentiating the modal sum, or repeating the cutoff argument for the derivative, gives $\dot{\gamma} = \Theta(\gamma^{-\beta-1})$ under gradient flow. Integration yields (7.11).

For finite depth, stability along the top eigendirection restricts $\gamma < 2L_d/\lambda_1$. Choosing $\gamma = cL_d$ with fixed admissible $c > 0$ gives the upper bound $O(L_d^{-\beta})$ by the same cutoff argument. Conversely, every stable choice has $\gamma = O(L_d)$, so modes $j \gtrsim L_d^{1/\nu}$ retain a constant fraction of their energy and give the matching lower bound. A spectral rank- S projection and an idealized context revealing the first n modes leave the tails beyond S and n , respectively; (7.10) gives their stated orders. $\square$

Proof of Proposition 7.5. Since $\gamma = w^r$, gradient flow in w gives

$$\dot{\gamma} = rw^{r-1}\dot{w} = -r^2w^{2r-2}\mathcal{L}'(\gamma) = -r^2\gamma^{2-2/r}\mathcal{L}'(\gamma).$$

When $\mathcal{L}(\gamma) \asymp \gamma^{-\beta}$, this is $\dot{\gamma} \asymp \gamma^{1-2/r-\beta}$. Integration gives $\gamma^{\beta+2/r} \asymp t$, and substitution into the loss proves (7.13). $\square$

Derivation of the two-point formula. Cauchy’s functional calculus gives

$$p(M) = \frac{1}{2\pi i} \oint_{\mathcal{C}} p(z)(zI - M)^{-1} dz.$$

Insert this representation for both polynomial factors in the quadratic risk, use Fubini’s theorem, and take the trace cyclically. This gives (7.17). Balancing $S^{-\nu\beta}$ with $L_d^{-\beta}$ gives (7.15); balancing it with q^{2L_d} gives (7.16). The Heavy-Ball prefactor changes only the lower-order $\log \log S$ term.

E Proofs for the scaling laws

Proof of Theorem 8.1. Conditioning on Y_j , decompose

$$f_j - \hat{f}_{L,j} = f_j - \mathbb{E}[f_j | Y_j] + \mathbb{E}[f_j | Y_j] - \hat{f}_{L,j}.$$

The two terms are orthogonal in L^2 by the defining property of conditional expectation. Their variances are $r_{n,j}$ and

$$[1 - (1 - p_L(\zeta_{h,j}))\chi_j(t)]^2 v_{n,j}$$

for $j \leq M_h$. For $j > M_h$ the estimator is zero and the risk is λ_j . Summing independent modes proves (8.5). The h^{2p} perturbation follows from the assumed squared-risk stability of the represented modes. $\square$

Proof of Theorem 8.3. For the context term, regular variation and an integral comparison give

$$\begin{aligned} \sum_j \frac{\sigma^2 \lambda_j}{\sigma^2 + n \lambda_j} &\sim \int_0^\infty \frac{c_\lambda}{x^a + n c_\lambda / \sigma^2} dx \\ &= c_\lambda^{1/a} \left(\frac{\sigma^2}{n} \right)^{1-1/a} \int_0^\infty \frac{du}{1 + u^a}. \end{aligned}$$

The last integral is $\pi/[a \sin(\pi/a)]$. The width law is the standard tail asymptotic for a regularly varying sequence. Similarly,

$$\begin{aligned} \sum_j w_j e^{-2\gamma_j t} &\sim c_w \int_0^\infty x^{-c} e^{-2c_\gamma t x^{-b}} dx \\ &= \frac{c_w}{b} (2c_\gamma t)^{-(c-1)/b} \Gamma\left(\frac{c-1}{b}\right). \end{aligned}$$

Finally apply Theorem 6.1 and (6.7) modewise in (8.5); the represented posterior-mean variance has finite trace, which is absorbed into C . $\square$

Proof of Proposition 9.1. The objective (9.2) is a strictly convex quadratic in B , hence its unique minimizer is $\mathbb{E}_U H_U^{-1}$. Haar invariance implies $V(\mathbb{E}_U H_U^{-1})V^\top = \mathbb{E}_U H_U^{-1}$ for every orthogonal V , so the expectation is a scalar multiple of the identity. Its trace determines the scalar and gives (9.3). $\square$

F Comment on the nonlinear outer iteration

Justification of Proposition 10.1. Uniform spectral equivalence puts every symmetrically preconditioned H_k in one interval $[\mu, L]$. Theorem 6.1 or (6.7) therefore yields a depth depending only on the desired forcing tolerance and on L/μ , not on the mesh. Once (10.2) holds, the update is an ordinary inexact Gauss–Newton step. The asserted local conclusions are the standard contraction estimates obtained by combining Jacobian Lipschitz continuity, uniform coercivity, and the forcing sequence; the learned metric affects only the cost of reaching the prescribed inner residual. $\square$

References

- [1] Ekin Akyürek, Dale Schuurmans, Jacob Andreas, Tengyu Ma, and Denny Zhou. What learning algorithm is in-context learning? investigations with linear models. In *International Conference on Learning Representations*, 2023.
- [2] Blake Bordelon, Mary I. Letey, and Cengiz Pehlevan. Theory of scaling laws for in-context regression: Depth, width, context and time. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=qA42mWsnbl>.
- [3] Edoardo Calvello, Nikola B. Kovachki, Matthew E. Levine, and Andrew M. Stuart. Continuum attention for neural operators. *Journal of Machine Learning Research*, 26:1–52, 2025. URL <https://www.jmlr.org/papers/v26/24-0879.html>.
- [4] Shivam Garg, Dimitris Tsipras, Percy Liang, and Gregory Valiant. What can transformers learn in-context? a case study of simple function classes. *Advances in Neural Information Processing Systems*, 35:30583–30598, 2022.
- [5] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are RNNs: Fast autoregressive transformers with linear attention. *International Conference on Machine Learning*, pages 5156–5165, 2020.
- [6] Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023. URL <https://www.jmlr.org/papers/v24/21-1524.html>.
- [7] Zhihao Li, Di Xiao, Zhilu Lai, and Wei Wang. Neural preconditioning operator for efficient PDE solves. *arXiv preprint arXiv:2502.01337*, 2025. URL <https://arxiv.org/abs/2502.01337>.
- [8] Boris T. Polyak. Some methods of speeding up the convergence of iteration methods. *USSR Computational Mathematics and Mathematical Physics*, 4(5):1–17, 1964. doi: 10.1016/0041-5553(64)90137-5.
- [9] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006. URL <https://gaussianprocess.org/gpml/>.
- [10] Yousef Saad. *Iterative Methods for Sparse Linear Systems*. Society for Industrial and Applied Mathematics, 2 edition, 2003. doi: 10.1137/1.9780898718003.
- [11] Andrew M. Stuart. Inverse problems: A bayesian perspective. *Acta Numerica*, 19:451–559, 2010. doi: 10.1017/S0962492910000061.
- [12] Johannes von Oswald, Eyvind Niklasson, Ettore Randazzo, João Sacramento, Alexander Mordvintsev, Andrey Zhmoginov, and Max Vladymyrov. Transformers learn in-context by gradient descent. In *International Conference on Machine Learning*, pages 35151–35174, 2023.
- [13] Grace Wahba. *Spline Models for Observational Data*. Society for Industrial and Applied Mathematics, 1990. doi: 10.1137/1.9781611970128.